31338 Network Servers

Lab 1c: Windows Server Management in VMWare

Aims

1. Understand the initial configuration of Windows Server 2025.

Task 1: Startup

The supplied VMware image of Windows Server 2025 has already been installed. It is an evaluation version and does not include a license key, so it will operate only for a limited evaluation period.

Use VMware Player to start the Windows Server virtual machine. Refer to Lab 1a if you need to review how to start a VMware image.

One of the first differences from Linux is the startup sequence. Windows Server does not use Esc to display boot logging in the same way, so allow the startup process to complete.

Windows Server 2025 first loads the graphical interface and prompts for the Administrator password. Press *Ctrl+Alt+Delete* to display the password prompt. In a virtual machine, the host operating system may capture this key sequence. If that happens, use *Ctrl+Alt+Insert* instead. If the keyboard does not have a dedicated Insert key, use the VMware *menu* and select *VM -> Send Ctrl+Alt+Delete*.

Enter the following password for the Administrator account:

```
student123!
```

After login, the desktop and Server Manager Dashboard should appear. Select Local Server to view the Windows Server 2025 machine.

Review or update the following information:

1. Confirm that the time zone is set to (UTC+10:00) Canberra, Melbourne, Sydney. Change it if required.
2. Check the networks. Ethernet0 and Ethernet1 should have IPv4 addresses assigned by DHCP, with IPv6 enabled. Selecting the IPv4 address assigned by DHCP link opens the Network Connections configuration page.
3. Change the computer name and description to something more meaningful, such as *Andy*. For now, leave the workgroup as WORKGROUP. Windows may request a reboot after the name change. You may reboot immediately or after completing the initial review.
4. Review the updates section to see when updates were last installed or checked, and how Windows Update manages updates. A server differs from a workstation because unexpected automatic reboots can disrupt services.

Record the configuration changes and any other noteworthy settings in your journal.

Scroll down to review events, services, and other information about the local server. If the server has not already been rebooted after a name change, reboot it before continuing.

Task 2: Server Management

Server Manager provides both a summary of server configuration and status and a main console for managing many server functions. It can be used to:

1. View and update the server summary, including computer and security information.
2. View and update server roles.
3. View and update server features.
4. Access common resources, support links, and the Best Practices Analyzer.

The Tools menu at the top of Server Manager opens additional administration tools. These include tools to:

1. View diagnostic information, such as Event Viewer and Device Manager.
2. View or update configuration through tools such as Task Scheduler, Windows Firewall, Services, and Computer Management.
3. Access backup and disk management tools.

Activities

1. Explore Device Manager. What hardware is installed on this machine? Does it look like the Linux image? Record your observations in your journal.
2. Identify the network address, or IP address, that connects the server to the network. Compare it with the Linux server. What are the subnet and gateway?
3. Add the Telnet Client and Simple TCP/IP Services features. These provide common TCP/IP services, including echo and daytime. Do not change Server Roles at this stage.

Check whether Simple TCP/IP Services is running by opening the Services panel and locating the Simple TCP/IP Services entry. If it is not started, determine how to start it by using the green arrow, the More Actions panel, or the right-click menu.

4. Test the services by opening Command Prompt from the Start menu under Windows System and entering:

```
telnet localhost 13
telnet localhost 17
```

Although PowerShell is useful, Command Prompt is preferable for this exercise because the telnet output is easier to observe.

5. From the host workstation, try the same commands against the server IP address. Can you connect? Why do you think this is the case?

Task 3: PowerShell

Almost all Windows Server 2025 configuration tasks can also be performed from the PowerShell. Some useful command families include:

`net`	contains functions for managing the local server
`netdom`	contains functions for managing a domain
`netsh`	contains functions for managing the local network

Use the following syntax to view command help:

```
net help command
```

Activities

1. View the services that are currently running:

```
net start
```

Observe the output. Is Simple TCP/IP Services running? It can be stopped with:

```
net stop "Simple TCP/IP Service"
```

Test whether the service has stopped, for example by using telnet. Then restart the service with net start.

2. Allow quote of the day through the firewall. The command should be entered as one line:

```
netsh advfirewall firewall add rule name="TCP Port 17" dir=in action=allow
protocol=TCP localport=17
```

Repeat the earlier telnet test from the host workstation to port 17. If the firewall rule is correct, the connection should now succeed.

Use your journal and appropriate documentation to study the syntax of these commands.

Task 4: Shutdown

Finally, shut down the virtual machine through the *Start* menu.

Windows Server requires a reason to be entered for shutdown. This is good administrative practice because it provides an audit record and helps the next administrator understand why the server was shut down.

Choose an appropriate topic, such as Other (Planned), enter a clear reason, and shut down the system.

Hint: if work needs to be paused temporarily, use VMware's suspend feature from *VM -> Power -> Suspend*. This saves a snapshot of memory in a .vmem file and runtime details in a .vmss file.